

1. What This Manual Covers

This is a plain, click-by-click manual for **running and using** the Smart Expense & Budget API through its web interface. It assumes zero prior knowledge of the tools involved — it only tells you what to click, what to type, and what you should see at each step.

For an explanation of *how the code works internally*, see the companion document “**Project Deep Dive.**”

2. Starting the Project

Every time you want to use this project, do these two things:

1. Open Command Prompt inside the `expense_api` folder.
2. Activate the virtual environment and start the server:

```
venv\Scripts\activate
uvicorn app.main:app --reload
```

You'll know it worked when you see:

```
Uvicorn running on http://127.0.0.1:8000
```

Leave this window open — closing it stops your project. Then open your browser and go to:

```
http://127.0.0.1:8000/docs
```

This page is called **Swagger UI**. It is your entire application's control panel — every feature of the project is usable from this one page, with no extra tools needed.

3. Understanding the Page Layout

The page lists your API's features grouped into sections:

- **Auth** — register, login, logout
- **Categories** — create and list spending categories (Food, Transport, etc.)
- **Transactions** — add, view, edit, delete income/expense entries
- **Budgets** — set a monthly limit per category
- **Reports** — see your monthly income/expense/budget summary

Each row shows a colored method tag:

Color	Method	Meaning
Green	GET	Fetch/view data
Blue	POST	Create something new
Orange	PUT	Update something existing
Red	DELETE	Remove something

Click any row to expand it and see its details.

4. Step-by-Step: Using the Project for the First Time

Step 1 — Register an account

Open **POST /auth/register** → click **Try it out** → fill in the JSON box:

```
{
  "email": "you@example.com",
  "password": "yourpassword",
  "full_name": "Your Name"
}
```

Click **Execute**. A **201** response means your account was created.

Step 2 — Authorize

Scroll to the top and click the **Authorize** button (padlock icon). A form appears asking for **username** and **password** — enter the same email and password you just registered with, leave **client_id** / **client_secret** blank, and click **Authorize**, then **Close**.

From this point on, every request you make on this page is automatically sent on your behalf — you don't need to log in again for the rest of the session.

Step 3 — Create a category

Open **POST /categories** → Try it out:

```
{ "name": "Food" }
```

Execute. Note the **id** returned (usually **1** for your first category) — you'll need it in the next step.

Step 4 — Add income

Open **POST /transactions** → Try it out:

```
{
  "type": "income",
  "amount": 30000,
  "description": "Salary"
}
```

Execute.

Step 5 — Add an expense against a category

```
{
  "type": "expense",
  "amount": 4000,
  "description": "Groceries",
  "category_id": 1
}
```

(Use the actual **id** of the category you created in Step 3.)

Step 6 — Set a budget for that category

Open **POST /budgets** → Try it out:

```
{
  "category_id": 1,
  "amount": 5000,
  "month": 9,
}
```

```
}
  "year": 2026
}
```

Step 7 — View your monthly report

Open **GET /reports/monthly** → Try it out → Execute (leave month/year blank to default to the current month). You'll get back a summary like:

```
{
  "month": 9,
  "year": 2026,
  "total_income": 30000,
  "total_expense": 4000,
  "balance": 26000,
  "budgets": [
    { "category": "Food", "budget": 5000, "spent": 4000, "remaining": 1000 }
  ]
}
```

This is the core output of the entire project.

5. Everyday Usage — What to Do Next

Once set up, here's how you'd use it going forward, in any order:

- **View all your transactions:** **GET /transactions**
- **Filter transactions:** **GET /transactions** supports optional filters — `category_id`, `type`, `start_date`, `end_date` — fill in only the ones you need, leave the rest blank.
- **Edit a transaction:** **PUT /transactions/{id}** — put the transaction's `id` in the path field, and only the fields you want to change in the body.
- **Delete a transaction:** **DELETE /transactions/{id}**
- **Add more categories:** repeat Step 3 for Transport, Shopping, etc.
- **Update a budget:** re-submit **POST /budgets** with the same category/month/year and a new amount — it overwrites the old one rather than duplicating it.
- **Check a different month's report:** **GET /reports/monthly** with `month` and `year` filled in.
- **Log out:** **POST /auth/logout** — this invalidates your current token, so you'd need to Authorize again to keep using the page.

6. Stopping and Restarting

To stop the server: click into the Command Prompt window and press **Ctrl + C**.

To start it again later, repeat Section 2 — your data isn't lost between restarts; it's saved permanently in your MySQL database. Only the *server process* stops, not your data.

7. Quick Troubleshooting

What you see	What it means
Browser says "site can't be reached"	The server isn't running — go back to Section 2
401 Unauthorized on any request	You're not authorized, or your token expired — click Authorize again
404 Not Found on a transaction/category/budget	You used an <code>id</code> that doesn't exist — check with the matching GET endpoint first
422 Validation Error	Something in your JSON body is missing, misspelled, or the wrong type — compare against the example

End of User Manual.